Background-contrastive neighbor embedding: method, evidence and open questions

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October 2026

1 Summary

This report describes the current state of the method, the reasoning and evidence behind each design decision, what it has been tested on, and what remains open.

The method. BC-NE embeds a target dataset X_T in two dimensions while suppressing variation that it shares with a background dataset X_B. Its core is a *background-suppressed neighbor graph*. For each target point, the method takes its nearest background points, subtracts their mean and their top few principal directions, and builds the k-nearest-neighbor (kNN) graph on what remains. The graph is then embedded with two-stage Neg-t-SNE, a negative-sampling neighbor embedding [3]. Optionally, the method adds *background negatives*: extra repulsion between target points whose background neighborhoods overlap. We write BC-NE for the method without background negatives and BC-NE+BN for the method with them.

Key findings.

1. **The residual graph is what makes the method work.** On MNIST-on-grass, a real-image benchmark of MNIST digits overlaid on grass photographs, BC-NE+BN beats the method's earlier form on every confirmation seed. That earlier form used background negatives on the raw kNN graph. Digit kNN accuracy is 0.973 vs 0.931 *(pre-registered)*. Without background negatives, the residual graph alone raises plain Neg-t-SNE by 0.10–0.13 *(descriptive)*.
2. **Background negatives become redundant once the graph is background-suppressed.** BC-NE+BN and BC-NE differ by at most 0.002 per seed, with one exact tie *(pre-registered)*.
3. **On MNIST-on-grass the method ties contrastive PCA (cPCA) followed by t-SNE, and is more stable across seeds.** Neither is consistently better *(pre-registered)*. Across seeds, BC-NE+BN ranges over 0.972–0.974 and cPCA→t-SNE over 0.941–0.979 *(descriptive)*.
4. **Global cPCA wins when the background is additive.** If target = signal + independent background, the difference of the target and background covariances is exactly the signal covariance *(our derivation)*. Accordingly, cPCA-based pipelines beat raw-graph background negatives on every additive benchmark we ran, synthetic *(descriptive)* and real *(pre-registered)*.
5. **A label-free "cluster, then cPCA" pipeline (local cPCA) beats raw-graph background negatives** on two synthetic benchmarks designed to defeat global projections: within-cluster signal accuracy 0.984–0.992 vs 0.914 *(pre-registered)*.
6. **On one of those benchmarks, raw-graph background negatives beat every global linear pipeline, including ones chosen with labels**: 0.914, against 0.530 for the best cPCA variant *(pre-registered)* and 0.624 for the best labeled projection *(descriptive)*.

Main open question. The residual graph has so far been tested only on MNIST-on-grass, whose background is additive, so that global cPCA already does about as well. Its intended advantage, suppressing a background that no single global projection can remove, has not yet been tested: neither on real data with a non-additive background, nor on the synthetic benchmarks where global cPCA fails (section 8).

How to read this report.

- Section 2 defines the problem, the datasets and the evaluation protocol. Everything after it uses these definitions.
- Section 3 reviews the published theory, together with one extension of our own.
- Section 4 specifies the method.
- Section 5 goes through the design decisions and the evidence for each.
- Section 6 collects theoretical observations about when linear suppression works.
- Section 7 presents the experimental results in full.
- Sections 8 and 9 list what is untested and what we propose next.

Every claim carries one of five labels: *(published)* for a cited result; *(our derivation)*; *(pre-registered)* for a verdict under a rule committed before the results existed; *(descriptive)* for a measurement with no verdict attached; and *(interpretation)* for our explanation, not directly tested.

2 Problem, data and evaluation

2.1 Problem and notation

We have a *target* dataset $X_T \in \mathbb{R}^{n_T \times D}$ and a *background* dataset $X_B \in \mathbb{R}^{n_B \times D}$ in the same feature space. The background contains only nuisance variation. The target contains the same kind of nuisance variation plus a target-specific *signal*. The goal is a two-dimensional embedding of the target points whose neighborhoods reflect the signal rather than the background. This is the neighbor-embedding analogue of contrastive PCA [1], which finds directions with high target variance and low background variance, i.e. the top eigenvectors of $C_T - \alpha C_B$ for a contrast parameter $\alpha \geq 0$.

Notation.

- **Embedding:** $y_i \in \mathbb{R}^2$ is the embedding of target point i, and $d_{ij} = \|y_i - y_j\|$. The Cauchy kernel is $\phi(d) = 1/(1+d^2)$, and $q_{ij} = \phi(d_{ij})$ is the model similarity.
- **Graph and sampling:** p_{ij} is the uniform distribution over the directed edges of the symmetrized kNN graph of the targets ($k = 15$). ξ is a noise distribution over pairs. m and m_B are the numbers of uniform and background negatives per edge.
- **Residual graph:** k_{loc} is the number of background neighbors used to form the residual, and r the number of local background directions removed.
- **Covariances:** C_T and C_B are the covariances of X_T and X_B, and C_S, C_G those of the signal and background components. Q^T is the transpose of Q.

2.2 Datasets

Synthetic benchmarks. All synthetic benchmarks share one template: $n_T = n_B = 5000$ points in $D = 50$ dimensions, with isotropic noise of standard deviation 0.1.

- **Background:** six equal-weight Gaussian background clusters, with centers on a regular simplex of edge 30 in a 5-dimensional latent space.
- **Signal:** four groups, with centers on a simplex of edge 6 in a 3-dimensional latent space, scaled by a strength ρ.
- **Choice of ρ:** the largest value at which plain Neg-t-SNE still hides the signal. A benchmark is *valid* only if plain Neg-t-SNE leaves signal kNN accuracy at or below 0.50 and background accuracy at or above 0.90, while the clean signal alone is separable (accuracy at least 0.95).

The benchmarks differ in how the signal and background combine (table 1). Labels for signal group and background cluster are used only for evaluation.

We call the cyclic and between-cluster benchmarks together the *subspace-conflict benchmarks*. In both, the background clusters are identical in X_T and X_B, so the population identity of section 6.1 still holds. The difficulty lies elsewhere: in the cyclic benchmark, keeping cluster k's signal block also keeps cluster $k-1$'s nuisance.

Table 1: The benchmarks used in this report.

name	construction	what it tests
additive (orthogonal, overlapping, nonlinear)	$x_T = \rho A z_s + B(z_b) + \varepsilon$, $x_B = B(z'_b) + \varepsilon$. Orthogonal: signal and background spans orthogonal. Overlapping: half the signal span inside the background span. Nonlinear: B a random tanh network. $\rho = 0.2, 0.2, 0.3$.	whether background negatives beat simple controls
weak-signal stress sets	the additive benchmarks at $\rho = 0.1$, where the signal is weak even in the high-dimensional kNN graph	what happens when the graph barely carries the signal
interaction	the signal includes $M(z_b, z_s) - M(z_b, s_0)$ for a random tanh network M, so its effect depends on the background state; 1 of 16 configurations was valid	a non-additive signal
cluster-specific signal	each background cluster has its own signal map	non-independent signal
cyclic	six 3-D blocks $Q_0, \ldots, Q_5$; in cluster k the signal lies in Q_k and the nuisance in Q_{k+1} (indices mod 6)	a case no single global projection can solve
between-cluster	the signal lies inside the between-cluster span U; the within-cluster nuisance lies in a separate block V	a case where cPCA needs $\alpha < 1$ and few components
MNIST-on-grass (real)	digits 0/1 at 25% intensity added to 28×28 grass patches; background = grass patches	real images, additive background

MNIST-on-grass. This follows the construction of the cPCA paper [1]:

- **Images:** MNIST digits 0 and 1 [8], at 25% intensity, added to random 28×28 grayscale patches cut from 1274 ImageNet grass photographs [5].
- **Target and background:** drawn from two disjoint halves of the photographs.
- **Seeds:** each seed has 5000 target and 5000 background images.
- **Recipe:** the authors' notebook, with five documented changes made for determinism and provenance.

The grass is added independently of the digit, so this dataset is additive.

2.3 Evaluation protocol

Pre-registration and seeds. Every experiment was pre-registered: its runs, decision rules and predictions were committed to the repository before the results they govern (section B). Each dataset is generated from a seed, which also seeds every run on it. Settings such as m_B are chosen on *selection seeds* 0–2. Verdicts use fresh *confirmation seeds*, usually 3–5. One confirmation seed failed the validity check and was replaced, under a rule added before any further result on that benchmark.

The verdict rule. Runs are paired by seed. "X beats Y, consistent across 3 paired seeds" means that the mean difference is positive *and* every per-seed difference is positive; anything else is reported as "not consistent". No p-values are reported. Reruns were checked to be bit-identical in every experiment, and runs were flagged for collapse or fragmentation (none was).

Metrics. All metrics are computed on the 2-D embedding.

- *Signal kNN accuracy*: a 15-nearest-neighbor classifier predicting the signal label, with stratified 5-fold cross-validation. On MNIST-on-grass this is *digit kNN accuracy*.
- *Within-cluster signal accuracy*: the same classifier, computed inside each true background cluster and size-weighted. It measures signal separation regardless of the global layout.
- *Detectability*: within-cluster accuracy computed on the high-dimensional data. It shows how much signal is there to recover.

- *Recall vs grass*: the fraction of a point's 15 embedding neighbors that are also among its 15 nearest images by grass patch. Lower is better.

Controls.

control	rules out that the gain comes from
plain Neg-t-SNE	(baseline)
uniform extra negatives (same number, drawn uniformly)	more repulsion
t-SNE-end spectrum reference ($\bar{Z} = 100n$, i.e. $c = 500/n$)	a shift along the attraction–repulsion spectrum
same-cluster oracle negatives (drawn from the true cluster)	knowing the background cluster
cPCA→Neg-t-SNE, cPCA→UMAP, cPCA→t-SNE	linear background suppression
t-SNE, PCA→t-SNE	standard practice

Compute. The synthetic experiments ran on a laptop CPU, and MNIST-on-grass on the McGill Mathematics Slurm cluster (CPU only). No comparison mixes the two platforms. On a benchmark rerun, they differed by at most 0.027 in within-cluster accuracy, below every verdict margin *(descriptive)*.

3 Background: neighbor embeddings as contrastive learning

t-SNE [13] and UMAP [10] can both be written as attraction along kNN edges plus repulsion between other pairs. In t-SNE the repulsion comes from normalizing q over all pairs; in UMAP it comes from sampled noise pairs *(published)* [3]. The balance between the two forces places a method on an attraction–repulsion spectrum, running from t-SNE-like to UMAP-like embeddings *(published)* [2].

Damrich et al. [3] make this precise through contrastive learning. Both estimators below sample positive pairs from p and noise pairs from ξ, and train a classifier to tell them apart.

- **Noise-contrastive estimation (NCE).** The classifier is $q_\theta/(q_\theta + m\xi)$. At the optimum $q_\theta = p$, whatever the full-support noise distribution *(published)* ([6]; Theorem 1 of [3]).
- **Negative sampling** [11, 9]. The classifier is $q_\theta/(q_\theta + 1)$; the noise probability does not appear in it. Corollary 2 of [3] shows that, with uniform noise over the $|X|$ pairs, negative sampling learns a rescaled distribution $q = (|X|/m)\, p$ *(published)*.

Neg-t-SNE is negative sampling with the Cauchy kernel. In the `contrastive-ne` package its classifier is $q/(q + c)$ for a constant c, with default $c = 1$, and moving c moves the embedding along the spectrum *(published)* [3].

Extension to a general noise distribution *(our derivation)*. Corollary 2 assumes uniform noise. Background negatives make the noise non-uniform, so we need the optimum for a general ξ. The expected negative-sampling loss with classifier $q/(q + c)$ is

$$\mathcal{L}(q) = \sum_x \Big[-p(x) \log \frac{q(x)}{q(x) + c} \; - \; m\,\xi(x) \log \frac{c}{q(x) + c} \Big], \tag{1}$$

where the sum runs over pairs $x = (i, j)$. Treat each $q(x)$ as a free variable and set the derivative to zero:

$$-\frac{p}{q} + \frac{p + m\xi}{q + c} = 0 \quad \Longrightarrow \quad q^*(x) = \frac{c\,p(x)}{m\,\xi(x)}. \tag{2}$$

For $p(x) > 0$ the loss tends to $+\infty$ both as $q \to 0$ and as $q \to \infty$, so this stationary point is the minimum. With uniform $\xi = 1/|X|$ and $c = 1$, eq. (2) recovers Corollary 2.

With background negatives, each edge with head i draws m uniform negatives and m_B negatives from $\xi_B(i, \cdot)$, the latter weighted by γ in the loss. The expected noise weight on a pair is then $m\,\xi_U(i, j) + \gamma\, m_B\, \tilde{\xi}_B(i, j)$, where $\tilde{\xi}_B(i, j) = p_i\, \xi_B(i, j)$ and p_i is the head marginal of p. The same calculation gives

$$q^*_{ij} = \frac{c\,p_{ij}}{m\,\xi_U(i, j) + \gamma\, m_B\, \tilde{\xi}_B(i, j)}. \tag{3}$$

Pairs that the background explains, i.e. with large $\tilde{\xi}_B$, receive a smaller target similarity, so they are pushed apart; this includes kNN edges whose endpoints are close for background reasons.

The derivation is idealized. It assumes:

- that each q_{ij} can be set freely, which a 2-D embedding cannot do exactly;
- that expectations replace sampling;
- that the package's in-batch uniform negatives are uniform over pairs;
- no clamping (the package clamps the classifier output to $[10^{-4}, 1]$).

UMAP and initialization. The loss that UMAP actually optimizes differs from the one stated in its paper, and has far less repulsion *(published)* [4]. Lemma 3 of [3] shows that UMAP is negative sampling with the model $1/d_{ij}^2$ in place of the Cauchy kernel. Its repulsion therefore diverges as $d \to 0$, while Neg-t-SNE's stays bounded *(published)*. Finally, we initialize every embedding from the first two principal components, scaled so that the first has standard deviation 1. This initialization is recommended for preserving global structure *(published)* [7, 3].

4 The method

Algorithm 1 summarizes the full pipeline. This section specifies each step.

4.1 Background-suppressed (residual) neighbor graph

For each target point x_i we find its k_{loc} nearest background points $N_B(i) \subset X_B$ by exact Euclidean search. Let μ_i be their mean, and $V_i \in \mathbb{R}^{D \times r}$ the top r right singular vectors of the centered neighbors $\{b - \mu_i : b \in N_B(i)\}$. The residual of x_i is

$$\tilde{x}_i = (I - V_i V_i^T)(x_i - \mu_i). \tag{4}$$

The attraction graph is the exact $k = 15$ nearest-neighbor graph of the residuals, symmetrized by union and with binary weights.

The defaults are $k_{\text{loc}} = 30$ and $r = 2$. They were selected on one dataset, and the choice barely mattered there (section 5.7). The residual is invariant to sign flips of the columns of V_i. It needs neither a clustering of the background nor a global projection. Only the graph changes: the initialization and the background negatives, if used, are computed on the raw features.

4.2 Two-stage Neg-t-SNE

We use Neg-t-SNE [3] as implemented in `contrastive-ne` v0.4.0, non-parametric, with the package defaults:

- $m = 5$ negatives per edge, drawn from the current minibatch;
- $c = 1$;
- minibatches of 2^{15} directed edges;
- SGD with learning rate 1, annealed linearly to 0;
- elementwise gradient clipping at ± 4.

Training is two separate fits on the same graph. Stage 1 runs 250 epochs from the PCA initialization. Stage 2 runs 500 epochs starting exactly from the Stage 1 embedding, with a fresh learning-rate annealing. Background negatives, when used, are active in Stage 2 only.

4.3 Optional background negatives

The background-negative distribution ξ_B pairs target points that look similar for background reasons.

1. **Background neighbors.** For each target point i, find its $k_B = 15$ nearest background points $N_B(i)$.
2. **Scores.** Score each pair of target points by the Jaccard overlap $s_B(i, j) = |N_B(i) \cap N_B(j)|/|N_B(i) \cup N_B(j)|$.
3. **Partners.** Keep the top $L = 30$ partners of each point. Ties with the 30th are kept up to $2L$; beyond that cap, a seeded random draw decides, never target distance.

4. **Distribution.** Set $\xi_B(i,\cdot) \propto s_B(i,\cdot)$. Target kNN edges are not excluded.

For each kNN edge with head i, Stage 2 adds m_B negatives $b \sim \xi_B(i,\cdot)$, with the loss term

$$-\gamma \sum_{b=1}^{m_B} \log\Big(1 - \frac{\phi_{ib}}{\phi_{ib} + c}\Big), \qquad \gamma = 1, \tag{5}$$

using the same c, kernel and clamping as the package's own negatives. A unit test confirms that one background negative and one uniform negative at the same distance contribute identical loss and gradient. By eq. (3), the term lowers the target similarity of pairs with overlapping background neighborhoods *(our derivation)*. Loosely, the embedding learns "target similarity divided by background similarity" *(interpretation)*.

Algorithm 1 BC-NE (with optional background negatives)

Require: targets X_T, background X_B; $k_{\text{loc}} = 30$, $r = 2$, $k = 15$; optional m_B

for each target x_i **do**
  $N_B(i) \leftarrow$ the k_{loc} nearest rows of X_B to x_i
  $\mu_i \leftarrow$ mean of $N_B(i)$; $V_i \leftarrow$ top-r right singular vectors of $N_B(i) - \mu_i$
  $\tilde{x}_i \leftarrow (I - V_i V_i^T)(x_i - \mu_i)$
end for
$G \leftarrow$ exact k-NN graph of $\{\tilde{x}_i\}$, union-symmetrized, binary
$Y_0 \leftarrow$ first two principal components of X_T, scaled so the first has std 1
$Y_1 \leftarrow$ Neg-t-SNE(G, init Y_0, 250 epochs)
if $m_B > 0$ **then**
  build ξ_B from Jaccard overlaps of the k_B-nearest background sets (raw features)
end if
$Y \leftarrow$ Neg-t-SNE(G, init Y_1, 500 epochs, plus m_B negatives per edge from ξ_B if $m_B > 0$)
return Y

4.4 Implementation

- **Package changes:** an 11-line hook in the vendored package lets the trainer add a term to each batch loss.
- **Graphs:** all kNN graphs are exact and built once per dataset. The package's own approximate graph is unseeded, so it is not used.
- **Determinism:** every fit runs with deterministic algorithms and seeded generators. The background sampler has its own generator, so with background negatives turned off the result is bit-identical to the unmodified package.

5 Design decisions and their evidence

This section takes each design decision in turn: what was chosen, the alternatives, and the evidence. The experiments behind the numbers are presented in full in section 7.

5.1 Neg-t-SNE rather than UMAP or t-SNE

The method needs an embedding whose optimum responds to the noise distribution. Negative sampling has this property, because the noise distribution is absent from its classifier (eq. (2)) *(our derivation)*. NCE does not: its optimum is p for any noise *(published)*. t-SNE has no noise distribution at all. We checked that the package's classifier uses a single constant c, not ξ *(descriptive)*. Among negative-sampling methods we rejected UMAP, because its effective $1/d^2$ model has a repulsion that diverges at short range *(published)*. Short range is exactly where background negatives act: on pairs that start close together because they share a background *(interpretation)*.

The choice has a cost. Without the background term, Neg-t-SNE scored below t-SNE on MNIST-on-grass: digit accuracy 0.858, against 0.884 for t-SNE and 0.891 for PCA→t-SNE *(descriptive)*. Comparisons with t-SNE-based controls therefore also change the embedding algorithm (section 8).

5.2 Putting the background into the noise distribution

The method's first version changed only the repulsion. It added background negatives on top of the raw kNN graph, which works only because negative sampling keeps the noise out of its classifier; under NCE the noise distribution would cancel at the optimum *(published)*. On the three additive synthetic benchmarks, this version beat the best uniform-extra-negatives control on every confirmation seed *(pre-registered)*, by a mean signal accuracy of +0.51 (orthogonal), +0.18 (overlapping) and +0.44 (nonlinear). It also beat the spectrum reference at every level *(pre-registered)* (table 2).

The uniform control is weak, however *(descriptive)*. With six equal, well-separated background clusters, about 5 in 6 uniform negatives land in another cluster, which is already far away, so they add little force within a cluster. Beating it shows that the repulsion must be targeted, not that ξ_B's targeting is the only kind that works. The next subsection uses a stronger control.

5.3 Background similarity from shared background neighbors

ξ_B pairs share their background but not their signal *(descriptive)*.

- **Synthetic data:** the two points of a pair belong to the same background cluster 100% of the time, and the excess of same-signal pairs over chance is at most +0.064.
- **MNIST-on-grass:** the same-digit excess is +0.055 to +0.074, and the grass patches of a pair are 3.8–4.0 apart, against 7.9–8.1 for random pairs.

The stronger control is the *same-cluster oracle*. It draws the extra negatives uniformly from the head's true background cluster, a deliberate use of labels. ξ_B beats it on the interaction benchmark (signal accuracy +0.39) and on both subspace-conflict benchmarks (within-cluster accuracy +0.21 and +0.22) *(pre-registered)*. So it is ξ_B's fine-grained neighborhoods that matter, not merely knowing the cluster *(interpretation)*. One caveat: on the subspace-conflict benchmarks the oracle was selected at the top of its grid ($m_B = 80$), and its selection mean was still rising slowly, by +0.01 to +0.02 from $m_B = 40$ to 80 *(descriptive)*.

5.4 The two-stage schedule

The package's built-in early-exaggeration phase does nothing at its negative-sampling defaults and does not reset the learning rate *(descriptive)*. Two separate fits instead give an exact, reproducible handoff: Stage 2 starts from the Stage 1 embedding bit for bit. The original reason for a warm-up was to let a global layout form before the background term acts, avoiding fragmentation. That is a design rationale, not a tested result: no single-stage ablation was run *(interpretation)*.

5.5 Moving the background into the graph

Background negatives change only the repulsion; the attraction still follows the raw kNN graph. Several measurements pointed to that graph as the limit *(descriptive)*:

- **Subspace-conflict benchmarks:** raw-graph background negatives reach within-cluster accuracy 0.91, about the detectability of the high-dimensional data itself (0.915–0.924).
- **Weak-signal stress sets:** where the signal is weak in the graph itself, the gains are small or absent.
- **Additive benchmarks:** none of about 925,000 kNN edges joins two background clusters, so attraction can never bring the clusters together.
- **MNIST-on-grass:** 44% of a target's raw-graph neighbors are also among its 15 nearest images by grass patch.

The "graph ceiling" is not strict, however. On MNIST-on-grass, raw-graph background negatives reached a digit accuracy of 0.931, above the raw graph's own digit accuracy on the same seeds (0.901) *(descriptive)*.

The main evidence for moving the background into the graph is therefore the residual-graph result itself. BC-NE+BN beats raw-graph background negatives on every MNIST-on-grass confirmation seed *(pre-registered)*.

5.6 A local residual graph rather than a global cPCA graph

The obvious alternative is to build the graph on the top 10 global cPCA components. On MNIST-on-grass, that graph was the cleaner of the two *(descriptive)*:

seeds	graph	graph digit accuracy	graph grass recall
selection (0–2)	global cPCA-10	0.970	0.006
selection (0–2)	residual ($k_{\text{loc}} = 30$, $r = 2$)	0.962	0.056
confirmation (3–5)	global cPCA-10	0.957	0.006
confirmation (3–5)	residual ($k_{\text{loc}} = 30$, $r = 2$)	0.962	0.057

On the confirmation seeds the accuracy order reverses, but cPCA-10 keeps the much lower grass recall. On additive data, then, the local graph has no clear advantage. Its case rests on backgrounds that are not additive, where no single global projection removes them; that case is untested (section 8) *(interpretation)*.

5.7 Defaults $k_{\text{loc}} = 30$ and $r = 2$

Before any residual graph was computed, we committed a graph-quality gate. On each selection seed, the selected variant had to beat the better of the raw-pixel and PCA-50 graphs by at least 0.02 in graph digit accuracy. It did, by +0.025 to +0.029 *(pre-registered)*.

The choice among variants barely mattered *(descriptive)*. All 15 variants on the grid, $k_{\text{loc}} \in \{15, 30, 100\}$ and $r \in \{0, 1, 2, 5, 10\}$, reach a mean graph digit accuracy of 0.954–0.964 (0.951–0.965 on individual seeds). Every one of them is above both reference graphs: raw 0.904 and PCA-50 0.935. Even $r = 0$, which only subtracts the local background mean, clears both (full grid in table 7). The labels used in selection therefore decided little, which supports a fixed, label-free default *(interpretation)*, although so far on this one dataset only.

5.8 Current status of background negatives

With the residual graph, background negatives were redundant on MNIST-on-grass *(pre-registered)*. The smallest value on the grid, $m_B = 1$, was selected, and accuracy fell slowly as m_B grew, from 0.973 to 0.968 at $m_B = 40$ *(descriptive)* (table 8). They have not been tested with the residual graph on any other data. A plausible reason for the redundancy *(interpretation)*: the residual graph and ξ_B use the same information, namely each target's nearest background points. ξ_B uses it for repulsion, the residual graph to define neighborhoods, and on this dataset putting it into the graph is what helps.

6 When linear suppression works: theoretical observations

6.1 The population identity behind cPCA

Claim *(our derivation)* (a standard argument). Suppose the target is $x_T = s + g + \varepsilon$ and the background is $x_B = g' + \varepsilon'$, where:

- s, g and ε are independent;
- g' has the same distribution as g;
- both noises have covariance $\sigma^2 I$.

Then $C_T - C_B = C_S$.

Proof. By independence the cross-covariances vanish, so $C_T = C_S + C_G + \sigma^2 I$ and $C_B = C_G + \sigma^2 I$. □

Consequences. The identity holds for *any* map producing g, linear or not, because only g's covariance enters and it cancels. At $\alpha = 1$, cPCA's top directions are therefore the signal's own directions, up to finite-sample error. For $\alpha > 1$,

$$C_T - \alpha C_B = C_S - (\alpha - 1)(C_G + \sigma^2 I),$$

so the signal survives in directions where the background has little variance *(our derivation)*.

Breaking independence is not enough to defeat this. Suppose each background cluster has its own signal map, with zero mean and lying outside the background's span. The cross-covariances still vanish, and the identity holds with a higher-dimensional C_S *(our derivation)*. The cluster-specific-signal benchmark was built this way, and it was stopped at calibration because cPCA→Neg-t-SNE already reached within-cluster accuracy 0.970–0.980 *(descriptive)*.

6.2 Finding vs cleaning

Claim *(our derivation)*. Let P be the orthogonal projection onto the k cPCA directions that are kept. Under the assumptions above,

$$\mathrm{Cov}(Px_T) = PC_SP + PC_GP + \sigma^2 P. \tag{6}$$

The identity guarantees that cPCA *finds* the signal directions. But the projected data are *clean* only if C_G has no variance in the chosen subspace. Otherwise the term PC_GP remains, and the embedding still sees that background variation.

Illustration *(descriptive)*: the between-cluster benchmark. For $\alpha < 1$, $C_T - \alpha C_B = C_S + (1-\alpha)(C_G + \sigma^2 I)$, so the top directions are first the between-cluster span U, which contains the signal, and then the nuisance block V.

- With $k \leq 5$ components, cPCA keeps only U and reveals the signal: within-cluster accuracy 0.989.
- With $k = 10$, the components also include V, and accuracy drops to 0.35.
- Embedding the labeled projection onto $U \cup V$ gives the same result, 0.34–0.36.

The between-cluster offsets that remain in U do not affect within-cluster accuracy.

6.3 Global vs local background variance

A local method can beat a global projection where the background varies strongly *globally* along the signal's directions, but weakly within a neighborhood *(interpretation)*. What decides whether cPCA fails, however, is the fraction of signal variance lying *outside* the directions of high global background variance, not the ratio of global to local variance itself *(descriptive)*. On the interaction benchmark's configurations, this ratio reached 46–69, but it was driven by a 2–10% tilt of the signal into the background's span. At least 87% of the signal variance lay outside that span on every seed, and cPCA used it.

6.4 The cyclic construction

In the cyclic benchmark, cluster k's signal lies in the block Q_k and its nuisance in Q_{k+1}, with nuisance standard deviation $\sigma_w = \sqrt{5/3}$ and signal scale $\rho = 0.2$.

Block-aligned projections *(our derivation)*. Consider projections onto unions of whole blocks. Revealing cluster k cleanly requires keeping Q_k and dropping Q_{k+1}. But keeping Q_k prevents a clean reveal of cluster $k-1$, whose nuisance is Q_k. Around the cycle of six, at most every other cluster can be revealed cleanly: 3 of 6.

General projections *(interpretation)*, not proven. Let a_k be the fraction of Q_k kept by a general projection. Revealing cluster k needs $\rho^2 a_k \gg \sigma_w^2 a_{k+1}$. Revealing two adjacent clusters would need a_k to fall by more than $(\sigma_w/\rho)^2 \approx 42$ per step, which the isotropic noise floor limits. We do not state this as a theorem.

Observation *(descriptive).* The best of the 64 labeled block subsets reveals exactly 3 clusters, with within-cluster accuracy 0.624. The best cPCA variants, which are not block-aligned, reach 0.530 on average (0.551 on the best seed).

Per-cluster cPCA *(our derivation).* Running cPCA within each cluster avoids the conflict. Within cluster k,

$$C_T^{(k)} - \alpha C_B^{(k)} = \rho^2 Q_k \Sigma_s Q_k^T - (\alpha - 1)\big(\sigma_w^2 Q_{k+1} Q_{k+1}^T + \sigma^2 I\big),$$

where Σ_s is the covariance of the signal latent. For any $\alpha \geq 1$ the top directions lie in Q_k, and each cluster gets its own projection. In the experiment, every per-cluster label-free α was above 1, ranging from 1.7 to 58.8 *(descriptive).*

6.5 The residual graph in these terms

(interpretation), untested. The residual of eq. (4) is a per-point version of the same idea: the local mean removes the cluster centroid, and the top-r local directions remove the local nuisance. In the cyclic benchmark each cluster's nuisance is 3-dimensional, so the default $r = 2$ removes only 2 of its 3 directions. On MNIST-on-grass, $r = 5$ was nearly identical to $r = 2$: graph digit accuracy 0.961 vs 0.962 at $k_{\text{loc}} = 30$ *(descriptive).*

7 Experimental results

All synthetic results in this section used the *raw* kNN graph with background negatives, the method's earlier form; the residual graph has not been tested on them. The real-data results (section 7.4) include both forms.

7.1 Do background negatives help beyond more repulsion?

On the additive benchmarks, yes. Table 2 shows large gains over every control at every level, consistent across seeds *(pre-registered).* Figure 1 shows the mechanism: each background cluster splits into its four signal groups, while the six clusters stay apart.

Table 2: Additive synthetic benchmarks: signal kNN accuracy on the confirmation seeds, mean [min, max]. Background negatives use the raw kNN graph at the m_B chosen on the selection seeds (10, 20 and 10). Every difference between background negatives and a control is consistent across 3 paired seeds *(pre-registered).* The last column, cPCA→Neg-t-SNE, is descriptive here.

background	plain	uniform (best)	spectrum ref.	background neg.	cPCA→Neg-t-SNE
orthogonal	0.312 [0.304, 0.322]	0.387 [0.374, 0.410]	0.346 [0.337, 0.352]	0.894 [0.890, 0.899]	0.989 [0.988, 0.990]
overlapping	0.368 [0.355, 0.388]	0.405 [0.391, 0.416]	0.384 [0.382, 0.387]	0.582 [0.573, 0.588]	0.980 [0.978, 0.982]
nonlinear	0.347 [0.339, 0.353]	0.429 [0.399, 0.464]	0.381 [0.364, 0.394]	0.868 [0.843, 0.902]	0.980 [0.976, 0.984]

The full pre-registered success criterion was met only on the overlapping level *(pre-registered).* The clause that failed elsewhere, "background accuracy drops", cannot be met by design. Background negatives add only repulsion, and no kNN edge joins two background clusters, so nothing pulls the clusters together *(descriptive).* Background accuracy stayed at 1.000 on the other two levels. The overlapping level's pass rests on about 10 boundary points.

The gain is not a change of scale *(descriptive).* Relative to plain Neg-t-SNE, background negatives spread the embedding by 1.11–1.25×, the best uniform control by 1.44–1.47×, and the spectrum reference by 1.79–1.87×. The two controls spread it more, yet recover little signal.

Against the stronger same-cluster oracle, ξ_B wins in every case *(pre-registered)*:

- interaction benchmark: by +0.385 to +0.388 per seed (0.858 vs 0.472), and still by +0.36 when the oracle uses $m_B = 40$;
- cyclic benchmark: by +0.19 to +0.23;
- between-cluster benchmark: by +0.22 to +0.23.

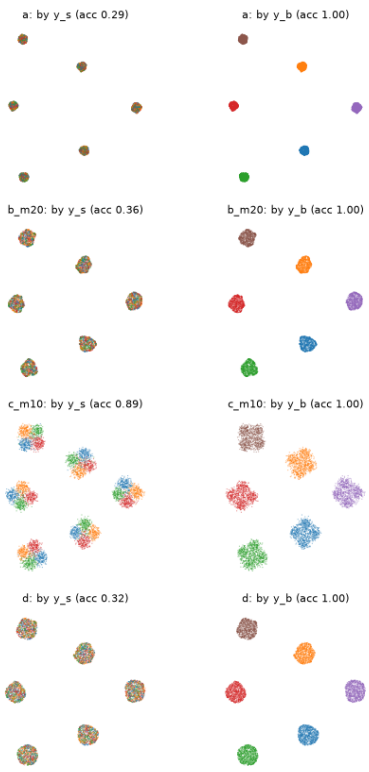


Figure 1: Orthogonal additive benchmark, selection seed 0, colored by signal group (left) and background cluster (right). Rows, top to bottom: plain Neg-t-SNE (`a`); 20 uniform extra negatives per edge (`b_m20`); 10 background negatives per edge on the raw graph (`c_m10`); spectrum reference (`d`). "acc" is kNN accuracy for the colored label.

On the weak-signal stress sets, which were run once and used for no choice, the gain over the best uniform control was small: +0.03, consistent on the orthogonal and nonlinear levels, and absent on the overlapping level *(descriptive)*.

7.2 When does global cPCA win?

Whenever the population identity holds and the signal lies mostly outside the background's high-variance directions.

- **Additive benchmarks** (table 2). cPCA→Neg-t-SNE reaches 0.980–0.989, beating raw-graph background negatives by +0.10, +0.40 and +0.11 *(descriptive)*. It also organizes the layout by signal across clusters: a fraction of 0.805–0.820 of a point's 15 embedding neighbors share its signal group but come from a different background cluster, against 0.000–0.001 for background negatives *(descriptive)*.
- **Interaction benchmark.** cPCA→Neg-t-SNE beats raw-graph background negatives, 0.984 vs 0.858 (per seed +0.086 to +0.155) *(pre-registered)*.
- **Choosing α.** The label-free rule was not a weak point on these benchmarks. After embedding, the label-free α and the label-chosen α differed by at most 0.003 in mean signal accuracy *(descriptive)*.

7.3 When do background negatives beat every global projection, and when does a local pipeline beat them?

Table 3 summarizes the subspace-conflict benchmarks, and fig. 2 shows the cyclic case.

Cyclic benchmark. Raw-graph background negatives beat the best cPCA variant by +0.36 to +0.41 per seed *(pre-registered)*. That variant was chosen per seed, with labels, over every α representative and $k \in \{3, 5, 10, 20\}$. They also beat the best labeled block projection, by +0.29 *(descriptive)*. Background negatives reveal all six clusters, with per-cluster accuracy at least 0.85; the best projection reveals three *(descriptive)*.

Between-cluster benchmark. Here the best cPCA variant (0.989) beats background negatives by 0.07–0.08 on every seed *(pre-registered)*. Label-free cPCA fails (0.255) on two counts *(descriptive)*. Its α rule picks $\alpha > 1$, which removes U along with the signal, and its fixed $k = 10$ brings in V (section 6.2).

Local cPCA. This baseline was added after the main results, with its predictions committed first. It works in three steps:

1. cluster X_B with k-means, choosing the number of clusters by silhouette, and assign each target to the nearest centroid;
2. within each cluster, run cPCA at that cluster's label-free α;
3. embed the per-cluster coordinates with a large cluster offset, so that no kNN edge crosses clusters.

Clustering found 6 clusters in perfect agreement with the truth (adjusted Rand index 1.0) on every seed. Local cPCA reached 0.984–0.992 on both benchmarks, at every $k \in \{3, 5, 10\}$, with label-free and true clusters alike. This confirmed the pre-registered predictions: at least 0.90, and label-free within 0.02 of true clusters *(pre-registered)*. On the cyclic benchmark it beat background negatives by +0.07 to +0.08 on every seed *(pre-registered)*. Discrete, well-separated background clusters are the most favorable case for this pipeline *(interpretation)*.

Table 3: Subspace-conflict benchmarks: within-cluster signal accuracy on confirmation seeds 3–5, mean [min, max]. "Best" rows are chosen per seed with labels, so they are upper bounds. Local cPCA uses label-free clusters and $k = 3$; the other k and the true-cluster versions are within 0.002 of it.

run	cyclic	between-cluster
plain Neg-t-SNE	0.334 [0.328, 0.344]	0.349 [0.346, 0.352]
uniform extra negatives ($m_B = 40$)	0.518 [0.503, 0.538]	0.522 [0.516, 0.529]
same-cluster oracle negatives ($m_B = 80$)	0.701 [0.689, 0.721]	0.691 [0.684, 0.699]
background negatives, raw graph ($m_B = 20$ / 10)	0.914 [0.912, 0.916]	0.914 [0.910, 0.921]
label-free cPCA→Neg-t-SNE	0.420 [0.377, 0.442]	0.255 [0.244, 0.262]
best cPCA variant (labels)	0.530 [0.500, 0.551]	0.989 [0.988, 0.992]
best labeled linear projection	0.624 [0.619, 0.628]	0.989 [0.988, 0.991]
local cPCA (label-free, $k = 3$)	0.989 [0.989, 0.990]	0.987 [0.984, 0.991]
detectability (high-D data)	0.919–0.924	0.915–0.924

7.4 Real data: MNIST-on-grass

Pre-checks *(descriptive)*. The digit signal is present in the raw kNN graph: digit kNN accuracy is 0.90 on the raw images and 0.93 after PCA-50, against 1.00 on the clean digits. But the raw graph's neighborhoods also follow the grass, with grass recall 0.44.

Raw graph. Raw-graph background negatives ($m_B = 5$) beat t-SNE [12], PCA-50→t-SNE and plain Neg-t-SNE on every seed *(pre-registered)*. In turn, cPCA→t-SNE (label-free α, 10 components) beats them on every seed *(pre-registered)* (tables 4 and 5).

Residual graph. Switching to the residual graph changes the picture.

- BC-NE+BN beats raw-graph background negatives, t-SNE and PCA→t-SNE on every seed *(pre-registered)*.

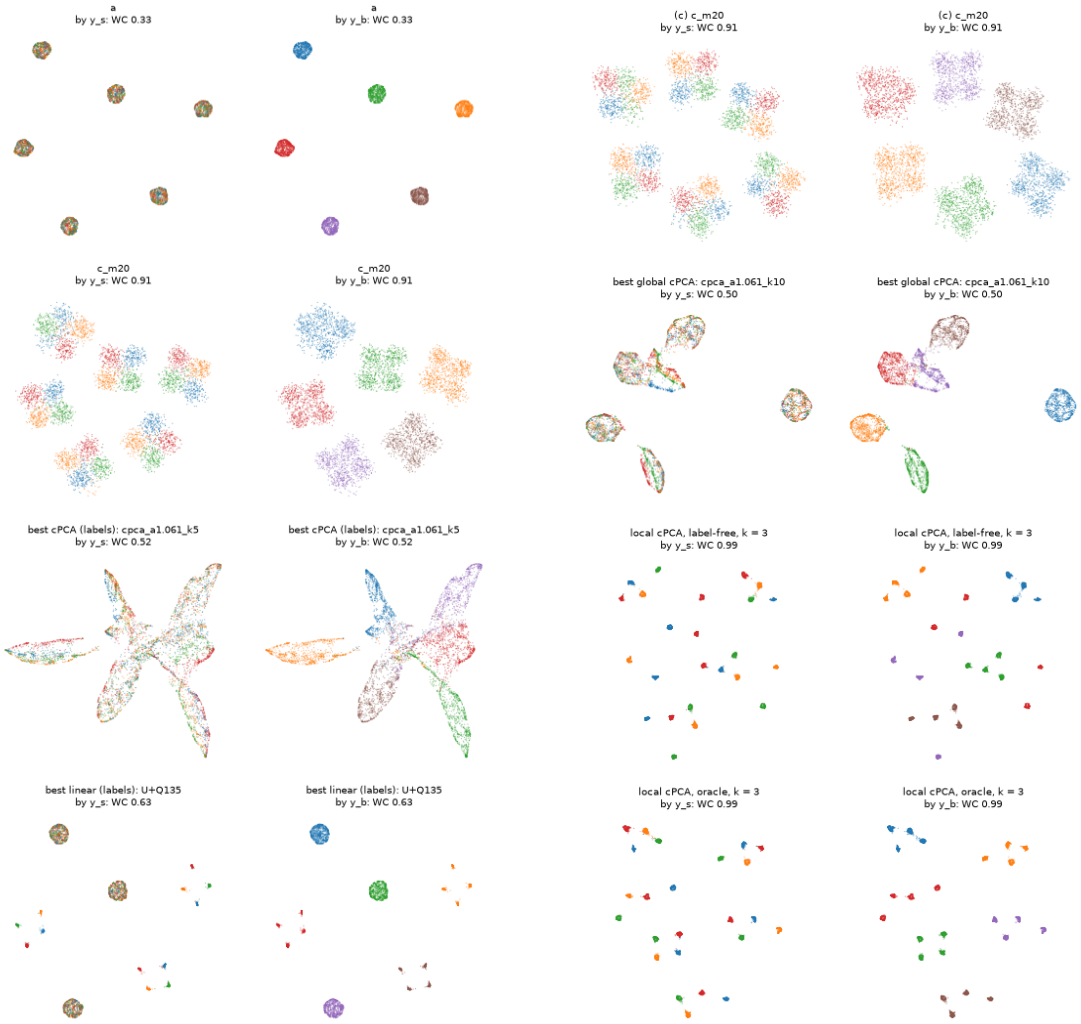


Figure 2: Cyclic benchmark. In each pair of columns, the left panel is colored by signal group and the right by background cluster; "WC" is within-cluster signal accuracy. **Left block (selection seed 0), top to bottom:** plain Neg-t-SNE (`a`); raw-graph background negatives, $m_B = 20$ (`c_m20`), which split every cluster into its four signal groups; the best cPCA variant chosen with labels; and the best labeled block projection (U plus blocks 1, 3 and 5), which reveals only clusters 1, 3 and 5. **Right block (confirmation seed 3), top to bottom:** raw-graph background negatives; the best global cPCA variant; local cPCA with label-free clusters; and local cPCA with true clusters ($k = 3$). Local cPCA gives 24 tight islands, one per cluster and signal group. The islands of one background cluster are not placed together, because the cluster offsets make all clusters equidistant.

- It ties cPCA→t-SNE, and it ties BC-NE without background negatives; neither comparison is consistent *(pre-registered)*.
- Recall vs grass falls from 0.022 to 0.003, the same as cPCA→t-SNE *(descriptive)*.
- In the embedding (fig. 3), the two digits form separate islands with no gradient in grass brightness. On the raw graph, by contrast, the digits split within one blob that still runs from dark to bright grass.

The BC-NE+BN seed-3 rerun was bit-identical. The raw-graph and t-SNE rows were reused from the earlier experiment, after checking that the code they depend on was unchanged and that every dataset matched its recorded checksum *(descriptive)*. The controls' embeddings are shown in fig. 4.

Table 4: MNIST-on-grass, confirmation seeds 3–5: digit kNN accuracy (mean [min, max] and per seed) and recall vs grass (lower is better). No run was flagged.

run	digit acc.	seed 3	seed 4	seed 5	recall vs grass
BC-NE+BN ($m_B = 1$)	0.973 [0.972, 0.974]	0.973	0.974	0.972	0.003
BC-NE (no background negatives)	0.972 [0.970, 0.974]	0.971	0.974	0.970	0.004
raw graph + background negatives ($m_B = 5$)	0.931 [0.926, 0.938]	0.926	0.938	0.930	0.022
plain Neg-t-SNE (raw graph)	0.858 [0.838, 0.871]	0.838	0.871	0.866	0.079
t-SNE	0.884 [0.872, 0.891]	0.872	0.891	0.890	0.135
PCA-50→t-SNE	0.891 [0.883, 0.898]	0.883	0.898	0.893	0.131
cPCA→t-SNE	0.955 [0.941, 0.979]	0.941	0.979	0.943	0.003

Table 5: MNIST-on-grass, pre-registered comparisons on digit kNN accuracy, confirmation seeds. Differences are the first run minus the second.

comparison	per-seed difference	mean	verdict
BC-NE+BN vs raw graph + BN	$+0.046, +0.036, +0.041$	$+0.041$	BC-NE+BN better, consistent
BC-NE+BN vs cPCA→t-SNE	$+0.032, -0.006, +0.028$	$+0.018$	not consistent
BC-NE+BN vs t-SNE	$+0.101, +0.083, +0.081$	$+0.088$	BC-NE+BN better, consistent
BC-NE+BN vs PCA→t-SNE	$+0.090, +0.076, +0.079$	$+0.082$	BC-NE+BN better, consistent
BC-NE+BN vs BC-NE	$+0.001, +0.000, +0.002$	$+0.001$	not consistent
raw graph + BN vs t-SNE	$+0.054, +0.046, +0.040$	$+0.047$	raw graph + BN better, consistent
raw graph + BN vs PCA→t-SNE	$+0.043, +0.040, +0.038$	$+0.040$	raw graph + BN better, consistent
raw graph + BN vs cPCA→t-SNE	$-0.015, -0.042, -0.013$	-0.023	cPCA→t-SNE better, consistent
raw graph + BN vs plain Neg-t-SNE	$+0.088, +0.066, +0.065$	$+0.073$	raw graph + BN better, consistent

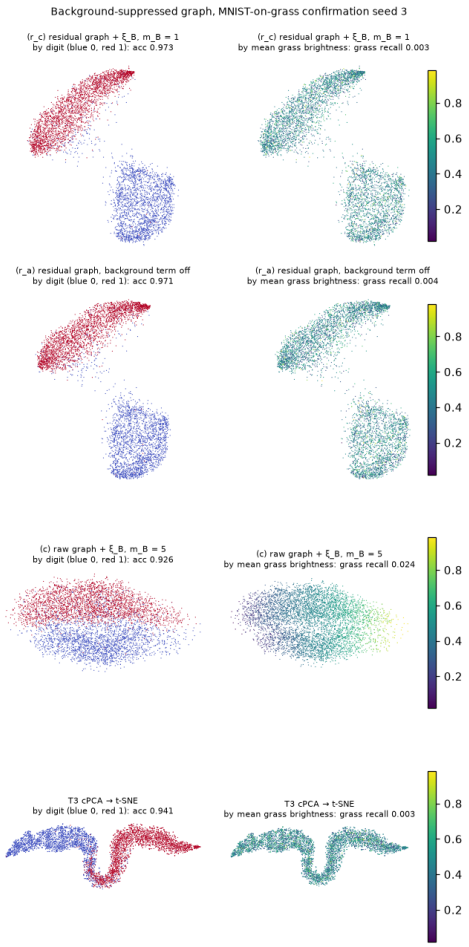


Figure 3: MNIST-on-grass, confirmation seed 3, colored by digit (left; blue 0, red 1) and by the mean brightness of each image's grass patch (right). Rows, top to bottom: BC-NE+BN (`(r_c)`); BC-NE without background negatives (`(r_a)`); raw graph + background negatives (`(c)`); cPCA→t-SNE (`T3`).

8 What has and has not been tested

Table 6 lists every experiment and its outcome for the method in the form tested.

Table 6: Every experiment and its outcome. "Raw" is the raw kNN graph with background negatives; "residual" is BC-NE / BC-NE+BN.

data	background	graph	competitors	outcome
3 additive synthetic	additive, discrete clusters	raw	uniform, spectrum ref., cPCA	beats controls; cPCA better
weak-signal stress sets	additive, signal weak in graph	raw	uniform, spectrum ref.	small gain on 2 of 3 levels
interaction	signal depends on background	raw	cPCA, same-cluster oracle	cPCA better; beats oracle
cluster-specific signal	cluster-specific signal maps	—	cPCA	stopped: cPCA already solves it
cyclic	subspace conflict	raw	cPCA, linear, oracle, local cPCA	beats all global linear; local cPCA better
between-cluster	signal in between-cluster span	raw	cPCA, linear, oracle, local cPCA	best cPCA and local cPCA better
MNIST-on-grass	real, additive	raw	t-SNE, PCA/cPCA→t-SNE	beats t-SNE controls; cPCA better
MNIST-on-grass	real, additive	residual	the above, and raw	beats raw and t-SNE; ties cPCA

The main gaps, in rough order of importance:

1. **Non-additive real data.** Every real-data result is on an additive dataset, where global cPCA is exact in population. This is also where the local residual graph's case against a global cPCA graph must be made (section 5.6).
2. **The residual graph on the subspace-conflict benchmarks,** the designs where global cPCA fails or needs labels. One open point is whether the default $r = 2$ suffices against their 3-dimensional local nuisance (section 6.5).
3. **Continuous or curved backgrounds.** Every background so far consists of discrete, well-separated clusters, which also favors cluster-then-cPCA pipelines.
4. **The role of background negatives.** They are redundant with the residual graph on MNIST-on-grass and untested with it elsewhere, for example when the residual graph is imperfect.
5. **Label-free defaults beyond MNIST-on-grass.** k_{loc} and r have been checked on one dataset.
6. **The algorithm confound.** Comparisons with t-SNE-based controls also change the embedding algorithm, and without the background term Neg-t-SNE was below t-SNE on MNIST-on-grass (section 5.1).
7. **A single-stage schedule** has not been compared with the two-stage one.
8. **Nonlinear contrastive methods**, such as contrastiveVI [14], have not been compared.
9. **Scaling.** Only $n = 5000$ has been tested. The residual construction does one small SVD per target point.

9 Proposed next steps

1. **A real dataset with a non-additive background,** with BC-NE (with or without background negatives) as the main method, compared against cPCA→t-SNE, t-SNE and PCA→t-SNE. The dataset is to be chosen together. It should be one where the nuisance interacts with the signal, so that global cPCA's identity fails.
2. **Optional: the residual graph on the existing cyclic and between-cluster benchmarks,** against raw-graph background negatives and local cPCA. This needs no new data and tests local against global suppression directly.

3. **What background negatives are for.** Test whether they help when the residual graph is imperfect, for example with a mis-sized k_{loc} or r. If they never help, drop them from the method.

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A Map to the repository

The repository (section B) organizes the work by phases, each with its own detailed report and results directory, and uses short internal labels for runs. This table maps the names used here to those phases and files, for readers who want more detail. The report does not depend on them.

name in this report	internal label	detailed report	results directory
additive benchmarks (orthogonal / overlapping / nonlinear), uniform control, spectrum reference, weak-signal stress sets	easy / medium / hard; runs (b), (d); stress	`REPORT.md` (Phase 4)	`results/phase4/`
cPCA→Neg-t-SNE / UMAP on the additive benchmarks	(e2) / (e3)	`REPORT_phase5prep.md`	`results/phase5prep/`
interaction / cluster-specific-signal benchmarks; same-cluster oracle	Design J (t60_proj_g2) / Design K; run (o)	`REPORT_phase5a.md`	`results/phase5a/`
cyclic / between-cluster benchmarks; local cPCA	L-cyclic / L-between; local_labelfree, local_oracle	`REPORT_phase5b.md`	`results/phase5b/`
MNIST-on-grass, raw graph; t-SNE / PCA→t-SNE / cPCA→t-SNE	(c), (a); T1 / T2 / T3	`REPORT_mnist_grass.md`	`results/mnist_grass/`
BC-NE / BC-NE+BN; residual-graph grid	(r_a) / (r_c); Phase 1	`REPORT_bsgraph.md`	`results/bsgraph/`
plain Neg-t-SNE	(a)	all	—
raw graph + background negatives	(c)	all	—
background-negative distribution	ξ_B	`REPORT.md`	`results/phase2/`
within-cluster signal accuracy	WC	`REPORT_phase5a.md`	—

Package internals (the sampling scheme, the classifier constant, the schedule facts) are documented in `NOTES.md`, and the original plan in `Initial_Project_Plan.md`.

B Reproducibility

Code. The code is at `https://github.com/Cassiebastress/bg-contrastive-ne` (public). Every result file records its software environment and the commit it ran at.

Pre-registration commits. Each was made before the results it governs.

experiment	commit
additive benchmarks: protocol and decision rules	`dad1c09`
global metrics and thresholds, before any cPCA result	`23459ff`
interaction / cluster-specific benchmarks: predictions; pre-registration	`4336ea0`; `9ed1110`
cyclic / between-cluster: predictions; pre-registration	`04a1450`; `9c35a0b`
local cPCA predictions	`a791d9b`
MNIST-on-grass (raw graph): pre-registration; selection	`8f916ee`; `3dd613a`
residual graph: graph-quality gate; embeddings	`3f0fc27`; `42bd31c`

Environment.

- Python 3.12, PyTorch 2.14.1 (CPU), numpy 2.5.2, scipy 1.18.1 and scikit-learn 1.9.1.
- `contrastive-ne` v0.4.0 (upstream commit `d5f0b53`), vendored with an 11-line hook.
- openTSNE 1.0.4 [12] for the t-SNE controls.

Compute.

- Synthetic experiments: an Apple M3 laptop, 4 threads.

- MNIST-on-grass: the McGill Mathematics Slurm cluster (AMD EPYC 9474F, CPU only), 1 thread per run, 24 runs in parallel. Before use, the cluster was validated: two runs were bit-identical, and it differed from the laptop by at most 0.027 in within-cluster accuracy on a benchmark rerun.
- Determinism: deterministic PyTorch algorithms, fixed thread counts, seeded generators and exact kNN graphs, with a bit-identical rerun checked in every experiment.

C Additional tables and figures

Table 7: Residual-graph grid on MNIST-on-grass, selection seeds 0–2, mean [min, max]. Selected: $k_{\mathrm{loc}} = 30$, $r = 2$. The rule was the highest mean, with variants within 0.002 counted as tied and broken toward the smaller r, then the smaller k_{loc}. Edge purity is the share of graph edges joining same-digit images.

graph	graph digit acc.	edge purity	grass recall	clean-digit recall
$k_{\mathrm{loc}} = 15$, $r = 0$	0.960 [0.959, 0.961]	0.945 [0.943, 0.945]	0.084 [0.083, 0.084]	0.062 [0.060, 0.063]
$k_{\mathrm{loc}} = 15$, $r = 1$	0.959 [0.958, 0.960]	0.948 [0.947, 0.950]	0.066 [0.065, 0.068]	0.064 [0.061, 0.066]
$k_{\mathrm{loc}} = 15$, $r = 2$	0.961 [0.959, 0.962]	0.950 [0.949, 0.951]	0.057 [0.054, 0.059]	0.065 [0.063, 0.067]
$k_{\mathrm{loc}} = 15$, $r = 5$	0.961 [0.958, 0.962]	0.951 [0.950, 0.954]	0.049 [0.047, 0.051]	0.068 [0.065, 0.070]
$k_{\mathrm{loc}} = 15$, $r = 10$	0.958 [0.955, 0.961]	0.951 [0.949, 0.952]	0.046 [0.045, 0.049]	0.068 [0.066, 0.071]
$k_{\mathrm{loc}} = 30$, $r = 0$	0.959 [0.956, 0.961]	0.941 [0.938, 0.942]	0.096 [0.093, 0.098]	0.060 [0.057, 0.062]
$k_{\mathrm{loc}} = 30$, $r = 1$	0.960 [0.959, 0.962]	0.946 [0.944, 0.947]	0.073 [0.072, 0.073]	0.063 [0.059, 0.065]
$k_{\mathrm{loc}} = 30$, $r = 2$	0.962 [0.961, 0.963]	0.950 [0.949, 0.951]	0.056 [0.054, 0.058]	0.066 [0.063, 0.068]
$k_{\mathrm{loc}} = 30$, $r = 5$	0.961 [0.960, 0.962]	0.953 [0.952, 0.954]	0.046 [0.043, 0.049]	0.070 [0.067, 0.073]
$k_{\mathrm{loc}} = 30$, $r = 10$	0.958 [0.955, 0.960]	0.951 [0.950, 0.954]	0.045 [0.041, 0.049]	0.071 [0.068, 0.074]
$k_{\mathrm{loc}} = 100$, $r = 0$	0.954 [0.951, 0.955]	0.930 [0.926, 0.931]	0.125 [0.122, 0.130]	0.054 [0.050, 0.056]
$k_{\mathrm{loc}} = 100$, $r = 1$	0.958 [0.956, 0.960]	0.937 [0.934, 0.939]	0.097 [0.094, 0.101]	0.057 [0.054, 0.060]
$k_{\mathrm{loc}} = 100$, $r = 2$	0.963 [0.962, 0.964]	0.948 [0.947, 0.950]	0.061 [0.059, 0.064]	0.065 [0.062, 0.068]
$k_{\mathrm{loc}} = 100$, $r = 5$	0.964 [0.962, 0.965]	0.957 [0.955, 0.958]	0.043 [0.040, 0.047]	0.074 [0.069, 0.076]
$k_{\mathrm{loc}} = 100$, $r = 10$	0.959 [0.958, 0.959]	0.954 [0.953, 0.956]	0.044 [0.040, 0.047]	0.075 [0.071, 0.079]
raw pixels	0.904 [0.903, 0.905]	0.844 [0.837, 0.851]	0.439 [0.432, 0.447]	0.020 [0.019, 0.023]
PCA-50	0.935 [0.932, 0.938]	0.872 [0.867, 0.878]	0.337 [0.328, 0.348]	0.020 [0.019, 0.022]
global cPCA-10 (descriptive)	0.970 [0.947, 0.986]	0.956 [0.924, 0.978]	0.006 [0.006, 0.006]	0.065 [0.018, 0.103]

Table 8: Selection of m_B on MNIST-on-grass, selection seeds 0–2: digit kNN accuracy, mean [min, max]. The rule was the highest mean, with ties broken toward the smaller m_B; the selected values are in bold.

m_B	residual graph (BC-NE+BN)	raw graph + background negatives
0	0.972 [0.971, 0.974]	0.850 [0.818, 0.876]
1	**0.973** [0.971, 0.974]	0.923 [0.910, 0.931]
2	0.972 [0.971, 0.973]	0.935 [0.934, 0.936]
5	0.972 [0.969, 0.974]	**0.937** [0.936, 0.939]
10	0.970 [0.968, 0.972]	0.930 [0.928, 0.932]
20	0.969 [0.967, 0.971]	0.924 [0.919, 0.927]
40	0.968 [0.966, 0.969]	0.922 [0.919, 0.925]

Table 9: Additive benchmarks: pre-registered per-seed differences in signal kNN accuracy (background negatives minus control) on the confirmation seeds. All are consistent across 3 paired seeds *(pre-registered)*.

control	orthogonal	overlapping	nonlinear
plain Neg-t-SNE	$+0.589, +0.589, +0.568$	$+0.227, +0.197, +0.218$	$+0.504, +0.552, +0.507$
uniform, same m_B	$+0.543, +0.538, +0.541$	$+0.181, +0.169, +0.182$	$+0.472, +0.492, +0.464$
uniform, best m_B (20)	$+0.519, +0.489, +0.513$	$+0.181, +0.169, +0.182$	$+0.444, +0.439, +0.436$
spectrum reference	$+0.556, +0.547, +0.541$	$+0.206, +0.198, +0.189$	$+0.479, +0.508, +0.474$

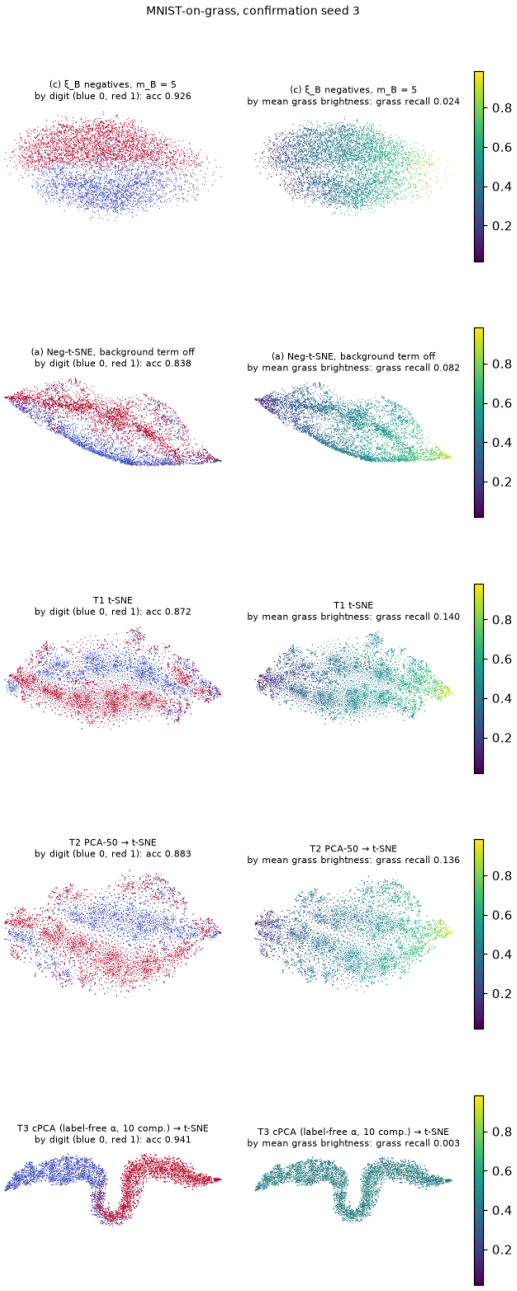


Figure 4: MNIST-on-grass controls, confirmation seed 3, colored by digit (left) and by grass brightness (right). Rows, top to bottom: raw graph + background negatives; plain Neg-t-SNE; t-SNE; PCA-50→t-SNE; cPCA→t-SNE. Plain Neg-t-SNE, t-SNE and PCA→t-SNE are organized partly by grass brightness, and their bright-grass end mixes the two digits.